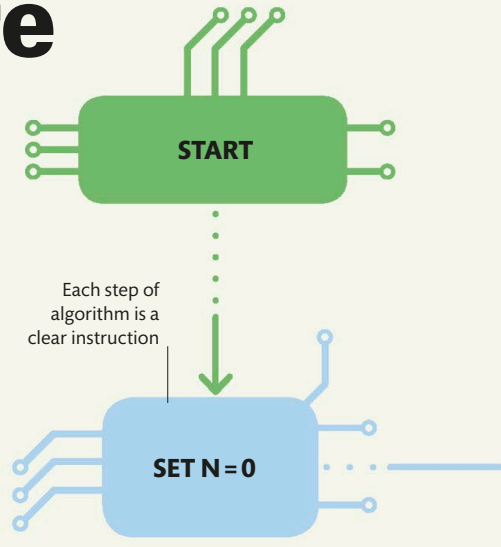
Computer software

The physical components of a computer are called hardware. Software is those parts you can't touch or hold—the programs, documents, sounds, and images. These exist as electric currents and charges that represent large collections of binary digits, 0s and 1s.



Algorithms and programs

An algorithm is a set of carefully worked out steps that achieve a specific task. A computer program is a collection of simple algorithms. A computer runs a program in order, but it may need to halt or jump to a different part of the program, depending on the input or the results of a calculation. It may also run a particular part of a program over and over, until a particular condition is met.

APPLICATIONS

An application is a program that a user launches on purpose, such as a word processing or photo-editing program. Applications can be launched by clicking with a mouse or trackpad, touching a smartphone screen, or using a voice command. Other programs are launched automatically by the operating system.



HOW MANY TASKS CAN A COMPUTER DO AT ONCE?

Many programs run concurrently, but a computer can execute only one instruction at a time, carrying out a small part of each program in turn.



Operating systems
An operating system is always running whenever a computer is switched on. A kernel, the core program that interacts with programs that are open, directs the inputs and outputs to wherever they are needed.